

Local Party Websites in Germany: A Chapter-Year Panel, 2015–2025

Data report for the public v1.0.0 release

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1 Summary

This dataset is a chapter-by-year panel describing the **observable web presence** of local political party organizations in Germany. It covers 2,611 local chapters (predominantly county-level Kreisverbände) of the seven main parties (CDU, CSU, SPD, GRÜNE, LINKE, AfD, FDP) in municipalities with a population of at least 5,000, observed annually from 2015 to 2025 (28,721 chapter-years). It is built from live crawls and Wayback Machine snapshots of chapter websites, from which structured measures are extracted: organizational presence, board composition, digital presence, mobilization infrastructure, issue emphasis, and a classified news layer (269,548 archived articles labeled by topic, function, and geographic focus). All content extraction and classification runs on a locally hosted

open-weight model, and every model-generated layer is human-validated. The public release ships chapter-year aggregates and coverage metadata only: no personal data and no reproduced website text.

Interpret the dataset as a panel of observable local party web presence and website-derived organizational signals, not as a census of local party activity, personnel, or issue positions. Whether a chapter-year is observed at all depends on website existence and Wayback archival coverage, which are themselves correlated with party, municipality size, and year (see Coverage). Every substantive field is a website-derived proxy: issue fields measure salience, not position; link and social flags reflect a site’s template, not verified activity; and the officials fields are website-listed, name-inferred aggregates.

2 Background

Local party organizations select candidates, recruit members, and run campaigns, but studying what they do and say at scale is hard because local manifestos and candidate lists are rarely digitized. Most studies therefore substitute national or state party positions for local ones. This dataset takes a different route: it measures what local chapters put on their own websites, and it does so longitudinally by reconstructing each site from archived snapshots. The measures map onto established website-coding frameworks (the five party-website functions of Gibson and Ward 2000) and issue families crosswalk to the Comparative Agendas Project (CAP) topic scheme, so the data connect to existing research.

3 Data collection and construction

The pipeline has five phases. (1) Roster discovery builds the chapter universe as 2,394 party-by-Kreis cells and matches them to the Destatis Gemeindeverzeichnis (municipality register); because the frame is party-by-Kreis, the unit is predominantly the **Kreisverband** (county-level party organization), and sub-county Ortsverbände are not systematically covered. (2) Site mapping crawls each chapter’s site and classifies pages by function. (3) Content extraction reads pages into structured schemas (officials, issue mentions, about/contact). (4) Historical extension queries the Wayback Machine CDX index, selects one snapshot per known chapter URL per year (a structural-page-priority rule; see the Reproducibility appendix), and fetches the archived HTML (about 1.03 million archived pages). (5) A research layer normalizes the extracted records and aggregates them to chapter-year features and measures.

All content extraction and classification run on a **locally hosted open-weight model** (Qwen3-30B-A3B-Instruct with guided decoding, which constrains outputs to the target schema); no scraped page content was sent to commercial cloud APIs. Structured records (officials, issue mentions, about/contact) are extracted per page; news articles are additionally deduplicated (one record per unique article) and each is classified into a policy topic family, an article function (press release, event, position statement, campaign, council work), and a geographic focus (local, regional, national). Gender is **name-inferred** from first names, a standard method benchmarked at roughly 94% accuracy; it is binary, unknown names are retained (see `gender_coverage`), and no migration-background error is assessed. Issue topics are assigned by a dictionary with a model fallback into about 25 families crosswalked to CAP major topics; these capture issue **salience**, not position. All model-derived steps feed deterministic chapter-year aggregates; no free text is retained in the release. The exact prompts, schemas, model settings, dictionary, and CDX parameters are in the

code repository and summarized in the Reproducibility appendix.

3.1 Sampling frame and URL discovery

The frame of possible chapters is constructed top-down before any website discovery: the Destatis Gemeindeverzeichnis restricted to municipalities with at least 5,000 residents (2,943 municipalities, aggregating to 399 Kreise), crossed with the party scope rules (AfD, FDP, Grüne, Linke, SPD everywhere; CDU outside Bavaria; CSU in Bavaria) yields **2,394 party-by-Kreis cells**. Chapter websites were then located per cell through four channels in decreasing order of authority, recorded per chapter in `url_source`: official party directories (999 chapters), the SPD organizational roster (8), systematic URL-pattern probing (869), and search-engine gap filling with model-classified results (735; search queries contained only party and place names). Discovery found a site for 97.8% of cells; the unfound cells concentrate in the parties with the weakest local digital presence. One structural caveat matters for the historical panel: discovery ran against the live web in early 2026, so chapters whose sites went offline before then without replacement are missing entirely, adding survivorship selection to early panel years (disclosed, not yet quantified).

4 Data records

The release contains eight data files at two grains (one row per chapter, or one row per chapter-year). Identifiers are stable across files.

File	Grain	Rows	Cols	Size
<code>chapter_roster.csv</code>	chapter	2,611	15	0.4 MB
<code>chapter_year_panel_hapter.csv</code>	chapter-year	28,721	42	13.3 MB
<code>chapter_year_panel_hapter_enriched.csv</code>	chapter-year	28,721	71	7.3 MB
<code>chapter_year_topic_hapter.csv</code>	chapter-year-family	25,832	5	1.0 MB
<code>chapter_year_news_hapter.csv</code>	chapter-year	13,883	13	0.8 MB
<code>chapter_year_news_hapter_topic.csv</code>	chapter-year-family	85,238	5	3.6 MB
<code>chapter_year_provenance_hapter.csv</code>	chapter-year	28,721	19	2.0 MB
<code>topic_family_cap_hapter_walk.csv</code>	family	26	4	small

`chapter_year_provenance.csv` carries non-sensitive source-level metadata (snapshot counts, capture timestamps, source-page and page-type counts, snapshot-selection mode, extraction-success counts) so any aggregate can be audited without exposing text or personal data. The `coverage_and_quality/` folder adds coverage, missingness, and extraction-validation tables.

Variables fall into seven groups (full definitions in `CODEBOOK.md`):

- **Identity and geography.** Party, organizational level, municipality, Kreis, `kreis_id`, `ags` (municipality key), population, and `url_source` (how the URL was found).
- **Coverage and observability.** `observability_status` and companion booleans, plus `is_quarantined`, `analysis_scope_included`, `n_usable_years`, and the release coverage taxonomy on every panel row (`coverage_basis`, `release_coverage_bucket`, `release_coverage_tier`), which makes the release denominator reproducible from the panel alone. Every substantive feature in the base panel carries a status column distinguishing an

observed positive, an observed zero, coverage-present-but-not-extracted, and missing coverage, so an apparent zero is never conflated with a missing site.

- **Organizational presence.** `board_listed`, `official_count`, `issue_block_count`, `office_address_listed`.
- **Digital presence.** Chapter-specific social accounts (Facebook, Instagram, X/Twitter, YouTube, Telegram, TikTok) and `n_social_platforms`, with national party accounts filtered out.
- **Web activity and mobilization.** Archived page counts, a news-activity proxy, and seven keyword-based link flags (donation, join, newsletter, contact form, program, events, volunteer) operationalizing the Gibson-Ward functions.
- **Board composition and issue emphasis (enriched panel).** Gender composition, chair presence and gender, turnover, professionalization shares, an `officials_source` flag, plus issue-topic counts, diversity, the dominant family, and 15 wide `topic_<family>_present` flags.
- **News volume and content (panel + news files).** `news_count` (deduplicated archived articles, dated by publication date where extractable), and per chapter-year label shares in `chapter_year_news.csv`: geographic focus (local/regional/national) and article function (press release, event, position, campaign, council work), with per-family article counts in `chapter_year_news_topic.csv` for issue-attention dynamics.

5 Observability status and scope

`observability_status` is a deterministic ladder computed per chapter-year: `no_known_website` (no URL) → `no_archive_coverage` (no same-year snapshot) → `crawl_not_attempted` → `crawl_failed` (snapshots exist but none fetched successfully) → `crawl_succeeded_but_no_extractable_pages` → `crawl_succeeded_but_below_usable_threshold` → `extractable_and_usable`. A chapter-year is **usable** when it has at least three successful pages and at least one extractable page (`usable_content_threshold_met`). `analysis_scope_included` marks chapter-years in the target universe and not quarantined; `is_quarantined` marks chapters that failed a roster-integrity check (225 chapters). Of 28,721 chapter-years, 25,245 are in analysis scope; the 3,476-row gap is out-of-universe or quarantined chapter-years.

6 Coverage and the denominator

The population frame is **2,394 party-by-Kreis cells**; 2,341 have a discovered chapter, and 2,611 chapters were discovered overall (some Kreise host more than one). Of the 28,721 chapter-years, 25,245 are in analysis scope. The release denominator is **researcher-usable coverage: 16,866 chapter-years (66.8% of scope)**, composed of 14,199 (56.2%) that meet the full structured-coverage threshold (`coverage_basis = full`) and 2,667 (10.6%) with observed feature values from below-threshold or fallback-routed pages (`coverage_basis = feature_level`). Most of the remainder is irreducible: 7,083 chapter-years (28.1%) have no same-year archived snapshot, and only 2 remain classified as potentially recoverable. The full funnel and breakdowns are in `coverage_and_quality/coverage_summary.csv`, and each panel row carries its own classification (`release_coverage_bucket`, `release_coverage_tier`).

Coverage is strongly non-random. The full-threshold usable share rises from about 35% (2015) to 73% (2025) and varies sharply by party (SPD 74.7%, GRÜNE 69.1%, CDU 61.1%,

FDP 52.6%, CSU 46.7%, LINKE 44.2%, AfD 37.2%). A descriptive linear-probability model (`missingness_diagnostics.csv`) shows usable coverage rising with municipality population (coefficient on log population about +0.04) and over time. Because observability is entangled with party, capacity, and archival density, missing cells must not be treated as zeros: condition analyses on `observability_status` or `coverage_basis`, and use the coverage and missingness tables (and Figure 1) to assess selection. This is a dataset of *observable* web presence, and the observation process is itself informative.

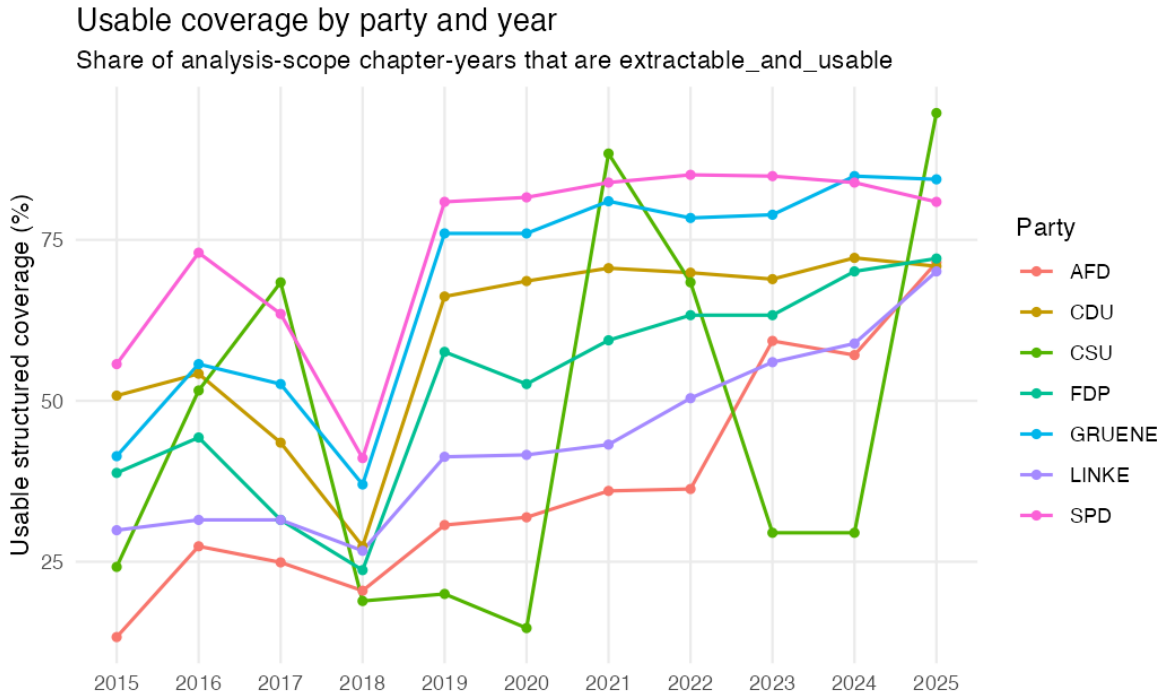


Figure 1: Usable structured coverage by party and year.

7 Technical validation

Three checks support data quality.

Human validation of every model-generated layer. All extraction and classification output was validated by a human coder reading extracted values next to source-page excerpts, in four stratified samples: the structured- extraction spot check (48 records across the officials / issues / about routes), the news-article extraction sample (63 articles: title, date, author, content), the issue topic-family sample (100 topic strings), and the news-label sample (77 articles, each with three labels: topic family, article function, geographic focus). Across all 288 reviewed records the human coder agreed with the extraction; per-layer counts are in `coverage_and_quality/validation_human_agreement.md`. The review sheets contain source-page excerpts and are therefore private; the public release reports only aggregate agreement.

External validity of the news labels. The share of news articles classified as election campaigning tracks the German electoral calendar: it peaks in federal and European election years (13.4% in 2024, 13.0% in 2025, 10.7% in 2021, 9.8% in 2019) and bottoms out in off-years (4.8% in 2015). This pattern is not built into the classifier and serves as an out-of-design validity check (see Figure 2).

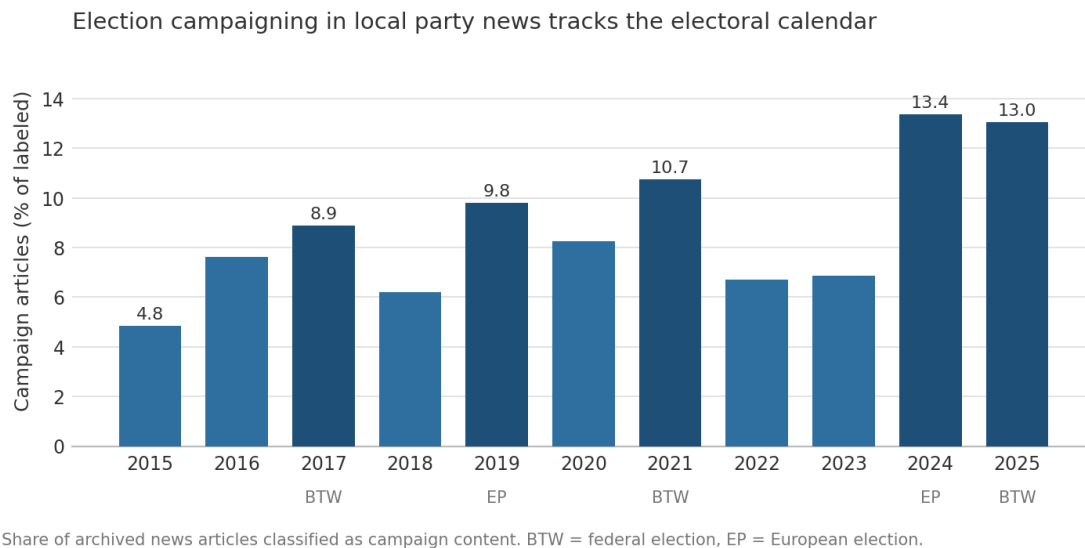


Figure 2: Campaign-content share of archived local party news articles by year. BTW = federal election, EP = European election.

Face validity. Name-inferred board female share orders as GRÜNE > LINKE > SPD > CSU > CDU > FDP > AfD; Telegram presence skews AfD; local-program pages skew AfD-high. Digital-presence flags exclude national party accounts. News content is predominantly locally focused (72% of articles), as expected for county-level chapter sites.

Construction integrity. Page-type classification carries no invalid labels, the bundle is regenerated by a single export script from the database, and the field set was reviewed dimension by dimension (relevance, likely accuracy, redundancy) with the resulting fixes applied. Issue-block extraction pools heterogeneous content (program pages, press items), so issue variables remain best treated as salience measures; the topic-family assignment itself is human-validated (100 of 100 sampled labels correct).

8 Usage notes and caveats

- Condition substantive fields on `observability_status`; filter to `analysis_scope_included = 1` to drop chapters that failed a roster-integrity check (`is_quarantined`).
- The extracted board pools internal party officers, candidates, and elected council representatives. `officials_source` flags the dominant source; condition board size, gender, and completeness measures on it.
- Issue fields measure **salience** (which topics a chapter emphasizes), not **position** (for or against). Topic-family classification is not separately ground-truth validated.
- Link and social flags reflect the site’s template, and “present” means a link appeared on an archived snapshot, not that the account was active that year.
- `news_count` counts deduplicated archived articles (publication-date year where extractable, snapshot year otherwise); it reflects Wayback archival density as well as posting volume. News label shares in `chapter_year_news.csv` are composition measures conditional on posting.
- Recommended filters: `analysis_scope_included = 1` plus either `coverage_basis = 'full'` (conservative) or any non-blank `coverage_basis` (full researcher-usable denominator,

with the basis as a robustness split).

- Read `ags` and `kreis_id` as strings; they are zero-padded and lose their leading zero under default integer inference.
- `has_wayback_coverage` and `usable_content_threshold_met` are exact recodes of `observability_status`; `n_social_platforms` is the sum of the six platform flags. Do not enter these alongside their components in one model.

8.1 Limitations

- **Selection into observability.** Chapters with usable websites differ systematically from those without (larger, more urban, party-skewed). The dataset measures the observable subset; it is not a random sample of local party activity.
- **Wayback coverage bias.** A chapter-year without a usable snapshot may reflect archival failure rather than absence of web activity, and high page counts can reflect crawler or site-structure behavior rather than real activity.
- **Temporal fragility.** A single annual archived snapshot may fall before or after a local election or a site redesign, so year-to-year change is noisy; the data are more reliable for broad cross-sectional web-presence measures than for fine annual dynamics.
- **Unit simplification.** Chapters are matched to one seat municipality and `ags`, simplifying a multi-municipality (Kreis-level) organizational geography; `org_level` is a rough label defaulting to `kreisverband`.
- **Non-random extraction error.** Model and dictionary errors may vary by template, party terminology, and period; see the validation study for measured rates.

9 Access and reuse

This public tier ships chapter-year aggregates and coverage metadata only. Person level records (names, emails, photos, biographies) and reproduced page text are withheld for data-protection reasons (GDPR) and are available to researchers on request under a use agreement, following the standard public-use versus restricted-use model. Public aggregate data files and release documentation are licensed under CC-BY-4.0; repository code is licensed under MIT.

10 Appendix: reproducibility

The full pipeline is in the code repository. Key artifacts:

- **Extraction prompts and schemas:** `src/extraction/extractors.py` (per-page-type prompts, applied verbatim by the local extractors in `scripts/hive/`) and `schemas.py` (record schemas). Model: Qwen3-30B-A3B-Instruct (open weights), temperature 0, guided/structured decoding; run on university HPC GPUs. No scraped page content was sent to commercial cloud APIs.
- **Classifiers and label sets:** `src/enrichment/classifiers.py` (name to gender, role to role category, topic to family), cached on normalized keys and classified with the same local model (`scripts/hive/cache_classify_vllm.py`); news labels via `scripts/hive/news_enrich_vllm.py`.
- **Issue dictionary and CAP crosswalk:** `config/issue_topic_crosswalk.csv`, `config/topic_family_cap_crosswalk.csv`.

- **URL discovery:** party directories, predictable-pattern probing, and Serper search classified by an LLM; the source is recorded per chapter in `url_source`. Historical URLs are found by querying the CDX index for each chapter’s known URL(s) plus archive hints, so URLs never linked from the discovered site may be missed (a survivorship risk).
- **Snapshot selection:** for each known URL and year, one snapshot is selected under a structural-page-priority rule (recorded per chapter-year as `selection_mode` in `chapter_year_provenance.csv`; see `src/wayback/` and the `wayback_url_selection_audit` table).
- **News layer:** `scripts/62/64/65` export, import, and clean the article layer (deduplication on chapter and normalized URL); `scripts/72/73` add the content labels.
- **Aggregation and export:** `scripts/20, 21, 55, 56` build the chapter-year layers; `scripts/52` builds the coverage taxonomy; `scripts/57` regenerates the release bundle deterministically; `scripts/58` builds the coverage/provenance diagnostics. Human validation was coded in the review app at `tools/validate/` and aggregates via `scripts/60`.

11 Citation

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11.1 Selected references

- Gibson, R. and Ward, S. (2000). A proposed methodology for studying the function and effectiveness of party and candidate web sites. *Social Science Computer Review*.
- Comparative Agendas Project. Master codebook (21 major topics).
- Santamaría, L. and Mihaljević, H. (2018). Comparison and benchmark of name-to-gender inference services. *PeerJ Computer Science*.